MATH 111 Calculus 1, NUU, FALL 2026
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Quiz 3 · Derivatives (2.2–2.6)

T10

15 minutes. 10 points. No notes, no calculators, no phones. Write your answers in the spaces provided and show all your work. You need not simplify your derivatives.

Name: _____

ID: _____

1	2	3	Total

Problem 1 *The rules.* **[4 points]**

Differentiate. Name the rule you use (power, product, quotient, chain) next to each answer.

(a) $y = 2x^5 + \dfrac{4}{x^2} - 6\sqrt{x}$

(b) $y = x^4 \cos x$

(c) $y = \dfrac{x^2 - 2}{x + 4}$

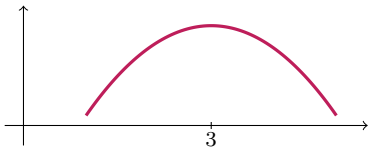
(d) $y = (x^4 + 3x)^3$

Problem 2 *From the lectures.* **[2 points]**

(Question 17 of Lecture 8, word for word.) Circle one letter, and give the reason in one or two lines.

The smooth graph of f below rises to a single high point at $x = 3$. Which statement about f' is true?

A. $f'(3)$ is the largest value of f'.
B. $f'(3) = 0$, and f' changes from positive to negative at 3.
C. $f'(3)$ does not exist.
D. $f'(x) > 0$ for every x.



Reason:

Problem 3 *Implicit differentiation.* **[4 points]**

The point $(2, 2)$ lies on the curve $2x^2 - xy + y^2 = 8$. Find y' by implicit differentiation, and then the equation of the tangent line to the curve at that point.

Show the differentiated equation, how you solve for y', the slope at the point, and the line.